State-Visitation Failures in On-Policy Distillation: An Audited Agent Case Study and Confirmatory Protocol

Anonymous ACL submission

Abstract

On-policy distillation supervises states visited by the student, but long-horizon agents may rarely reach persistent player states where a teacher has a measurable conditional advantage. We study a 2B-parameter language model acting for six hours through 17 typed tools in an open-source MMORPG. In one reported run per arm, natural-visitation distillation changes tool use and interaction tempo but leaves the quest score at 12/30. A second round augments student rollouts with hand-selected, verified intermediate player states. Evaluated from a fresh, unseeded world, the resulting policy scores 15/30; all three clustered prompt variants pass a wall that none passes under the reported base or first-round policy. This is a reported recovery-off observation, not a replicated causal estimate: the training rounds differ in more than state source. A targeted teacher-forcing probe also finds that malformed history can locally reverse the teacher's syntax preference. In rewritten runtime logs, correction markers do not repeat within a session. These observations motivate a confirmatory comparison in which training-only player states are selected by frozen student-visitation, teacher-advantage, validity, and recoverability criteria. Random-valid, progress-matched, teacher-success-prefix, natural-OPD, and guided-OPD controls separate targeted coverage from generic resets and prefix curricula. We present a falsifiable protocol and audited failure analysis, not an established method effect.

1 Introduction

Post-training a small language model to act over long horizons is not just a token-prediction problem. The model chooses which environment states become training data. If it never reaches a state where its teacher knows what to do, dense teacher feedback is unavailable exactly where it is needed. This creates a state-visitation bottleneck: improving token-level agreement on common states can alter style without transferring the competence that distinguishes the teacher.

We examine that bottleneck in a persistent tool-use setting. A Qwen-family 2B student plays a tile-based MMORPG through a typed action interface. The world persists across conversation rollovers, and benchmark progress requires long chains of gathering, navigation, dialogue, and quest actions. A scaffolded 4B teacher reaches states that the 2B student does not. We apply reverse-KL on-policy distillation to the student's own emitted tokens. The historical comparison contrasts natural student visitation with a mixture that begins some rollouts from verified intermediate database states; the confirmatory protocol additionally freezes a measurable state-selection rule before training.

The present evidence is hypothesis-generating and under-replicated. It contains one complete six-hour run per principal arm; the three agents within a run are prompt variants sharing weights, harness, and launch conditions, not independent experimental units. We therefore separate observations from claims throughout. The observed sequence is: base 12/30, natural-visitation round 12/30, and state-augmented round 15/30. In the last arm, every prompt variant passes the same wall from an unseeded evaluation world. This supports the hypothesis that training visitation matters, but it does not isolate state seeding from all round-to-round changes or estimate population-level uncertainty.

This working draft makes three bounded contributions:

1. We audit a state-visitation failure case and

separate observations that survive artifact review from causal claims that do not. We provide a confirmatory protocol with controls capable of falsifying the proposed player-state intervention.

2. We distinguish behavioral imitation from competence in the surviving historical record: the first round closely changes measured tool mix and tempo toward the teacher while the benchmark score remains fixed.

3. We document an interface failure that token-average analyses obscure: in a targeted teacher-forcing probe, malformed history reverses the teacher's local preference. This secondary diagnostic is separated from a runtime recovery result and from any general mechanism claim.

The broader lesson is methodological. Model weights, visited states, and runtime affordances are different interventions. A credible agent-training result must preserve and cross them rather than roll them into one headline.

2 Setting

2.1 Persistent tool-use benchmark

The environment is Kaetram, an open-source tile-based MMORPG backed by a game server and MongoDB. Each policy controls three agents with different priority prompts (grinder, completionist, and explorer) through 17 typed tools. Tools cover observation, navigation, combat, gathering, dialogue, quest inspection, inventory use, and crafting. The prompt includes procedural quest guidance, so the benchmark measures execution of a supplied plan rather than autonomous world-model discovery.

The primary development metric, Core-3, counts newly reached stages in three quest chains: Foresting (3 stages), Herbalist's Desperation (3), and Rick's Roll (4). Each agent can gain at most ten stages; a three-agent run therefore scores in $[0, 30]$. Runs begin from a freshly initialized world and continue for six hours. Conversation sessions roll over under a fixed context budget while game state persists. The intended confirmatory experimental unit is a full run from a fresh world. Prompt variants are clustered observations within that run.

2.2 Agent and interface contract

The policy acts in an observe–reason–act loop through an OpenAI-compatible model endpoint and a typed tool server. Navigation and selected interactions contain deterministic affordances such as path finding and automatic walking; the model still decides goals, tools, and arguments. The current development record contains an important interface asymmetry: older base-model runs used a native model-visible tool schema while some fine-tuned runs did not receive an identical serialization. We therefore do not use those comparisons as primary evidence. The confirmatory implementation requires normalized full-schema hash parity across training and every evaluation arm.

3 On-Policy Distillation Under Visitation Constraints

Let $x \in \mathcal{X}$ denote persistent player state (including position, inventory, skills, and quest progress) and $h \in \mathcal{H}$ the model-visible interaction history. The joint occupancy state is their joint value $z = (x, h)$; two rollouts with the same database snapshot but different visible histories need not induce the same action distribution. Let μ initialize player states, $q(h \mid x)$ initialize visible histories, and $d^{\pi}_{\mu,q}(x, h)$ denote the joint occupancy induced thereafter by policy $\pi(a \mid h)$ and the environment dynamics, whose transitions depend on the latent player and world state. The model does not observe x directly; it acts only from h, which may contain tool observations derived from x. The canonical serving initializers are μ_0 and q_0. A reverse-KL on-policy objective can be written

$$\mathcal{L}_{\text{OPD}} = \mathbb{E}_{(x,h)\sim d^{\pi_S}_{\mu,q}} \mathbb{E}_{a\sim\pi_S(\cdot|h)} \left[\log \pi_S(a \mid h) - \log \pi_T(a \mid h)\right]. \quad (1)$$

The teacher scores the student's emitted tokens. Training uses a clipped importance-weighted update around the rollout policy, with advantages frozen at data-build time. Full implementation details appear in Appendix A.

Equation 1 makes the bottleneck explicit. If a teacher advantage is concentrated on a set $H \subseteq \mathcal{X} \times \mathcal{H}$ for which $\Pr_{(x,h)\sim d^{\pi_S}_{\mu_0,q_0}}[(x, h) \in H] \approx 0$, token-dense grading supplies almost no information about that advantage. We change the state source, not the loss: some student rollouts begin from verified intermediate player states, and their records are mixed with natural student rollouts. The student still acts and the same teacher still grades every emitted token. Because changing x can also change the compatible h, the confirmatory design crosses player-state initialization with matched history construction rather than treating a database snapshot as the complete policy state.

The seeded states are typed persistent player states rather than textual demonstrations. Player rows specify quest stage, skill level, inventory, and a verified walkable position. End-to-end loadability is necessary but not sufficient for legal reachability. Each confirmatory candidate must also carry either a hash-verified witness trajectory from the canonical start that is replayed by a pinned checker, or an invariant certificate executed by a pinned checker that establishes a valid path to the snapshot. Hash continuity or certificate-shaped metadata alone is not sufficient. Evaluation is always unseeded. This resembles reset-based curricula in reinforcement learning (Salimans and Chen, 2018; Resnick et al., 2018; Ecoffet et al., 2021). TCOD uses teacher-success prefixes to initialize multi-turn OPD curricula (Wang et al., 2026); Guided-OPD mixes teacher and student turns and anneals guidance to zero (Li et al., 2026); ReOPD treats prefix selection as a teacher-reliability problem (Liao et al., 2026). These works preempt generic novelty for intermediate starts or state-distribution design. Our narrower object is a direct, prefix-independent player-state initializer. A candidate state need not lie on a successful teacher trajectory and is selected by a frozen rule rather than by progress alone.

Let $v(x)$ be estimated natural student visitation, and let $p_T(x, q)$ and $p_S(x, q)$ be repeated teacher and student success rates under a frozen history constructor q. The confirmatory selector admits a player state only if it passes player-state and compatibility invariants, is unfinished and task-relevant, is recoverable, satisfies $v(x) \leq \tau_v$, has certified reachability, and satisfies $p_T(x, q) - p_S(x, q) \geq \tau_\Delta$ together with a minimum teacher success threshold. Counts and denominators, source-run identifiers, complete snapshots, and hash-verified validation artifacts are frozen before outcomes. This rule distinguishes the proposed method from the hand-selected milestone used in the exploratory round.

4 Evidence Protocol

We grade evidence before interpreting it. A *run* is the independent unit; agents within a run share the policy, launch, infrastructure, and analysis pipeline. Exact scores from single runs are descriptive. Counts over agents or sessions diagnose mechanisms but cannot be treated as independent replications of a policy effect.

The principal comparison uses the same 2B base checkpoint, six-hour unseeded evaluation protocol, benchmark, and nominal harness. Round 1 trains on natural base-policy rollouts. Round 2 initializes from round 1 and mixes natural round-1 rollouts with 2,326 records collected by the round-1 policy from a verified Herbalist milestone state. The round-2 corpus contains 7,849 states, split into 7,024 training and 825 held-out records. Because round 2 also changes the initialization and data batch, round 1 versus round 2 is not a clean seeded-versus-unseeded training ablation.

The *wall* column measures passage of Herbalist stage 1 by the three prompt variants. It is a useful within-run consistency check, not a binomial sample of independent policies. We preserve raw pre-rewrite emissions for confirmatory format analyses; rewritten history is not an authoritative defect counter.

5 Exploratory Results

5.1 Natural-visitation OPD transfers behavior, not score

Round 1 leaves Core-3 unchanged at 12/30 and every agent reaches the same 4/10 stage profile as base. Nevertheless, several behavioral measurements move toward the 4B teacher (Table 2). The average number of turns per session exactly matches the teacher, observation and navigation shares approach the teacher, and query frequency overshoots it. The result is consistent with mode-seeking behavioral imitation, while showing that agreement on frequently visited states is insufficient for this quest wall.

Per-verb reverse-KL signs and behavioral movement are also aligned in the development

Table 1: Observed development runs and the claims they permit. Scores are descriptive because there is one run per arm. Recovery is a runtime interface affordance.

Policy	Training states	Recovery	Core-3	Wall	Interpretation
Base 2B	–	off	12	0/3	Reference run
OPD round 1	natural	off	12	0/3	Behavior changed; score did not
OPD round 2	natural + seeded	off	15	3/3	Weights-only, unseeded observation
Round-2 weights	–	on	17	–	Recovery ablation; one run
Round-3 weights	broader seeded	on	18	–	Weights plus interface; not pure weights

Table 2: Behavior changes in the first OPD round. Percentages are shares of tool calls in one run per policy; they are diagnostics, not independent trials.

Measure	Base	Round 1	Teacher
Turns per session	9.1	**6.1**	6.1
Observe share	26.6%	**35.5%**	36.1%
Navigate share	14.3%	**9.4%**	9.1%
Attack share	17.9%	**6.6%**	9.1%
Query-quest share	5.3%	**18.0%**	8.7%
Chars./turn	411	**1,248**	864

logs: verbs whose held-out divergence falls generally move toward the teacher's usage, while query-quest both diverges and behaviorally overshoots the teacher. This is descriptive triangulation. It does not establish that a specific token update caused downstream quest behavior.

5.2 The seeded-training round is followed by unseeded wall passage

The 4B teacher demonstrates competence beyond the Herbalist stage-1 wall. In the reported base and round-1 runs, zero of three prompt variants crosses that wall; the available record does not provide a defensible denominator for a broader natural-visitation rate. Seeded collection starts the round-1 policy at the accepted quest with the required gathering skill and near the target resource. All three seeded collectors pass the wall, creating student-authored actions and teacher scores in this region.

After training on the natural-plus-seeded mixture, round 2 is evaluated from a fresh unseeded world with recovery disabled. It scores 15/30, with each prompt variant at 5/10. All three pass Herbalist stage 1, compared with zero of three under both the base and round-1 policies. Logs show earlier quest acceptance, explicit tracking of the skill gate, resource rotation after empty harvests, and less premature return to the quest giver. These are concrete trajectory differences consistent with the intended mechanism.

The valid statement is deliberately narrow: *a seeded-training round was followed by unanimous unseeded wall passage in one run.* The present data do not show that seeding alone caused the improvement. The required causal experiment trains fresh students from the same checkpoint under identical budgets, seeds, and data construction, varying only whether the verified state mixture is included (Section 7).

5.3 Weights and recovery solve different failures

Round 3 combines broader state seeding with a runtime recovery affordance and scores 18/30. That number is not a pure-weights result. Applying the same recovery affordance to round-2 weights produces 17/30 in one run, suggesting that most of the round-2-to-round-3 stage difference may come from the interface. The reported recovery-off contrast is base 12/30 versus round 2 15/30, one run per arm; exact checkpoint, render, seed, and artifact parity is unavailable, so even this contrast is not a clean causal weights comparison.

The recovery mechanism detects a dropped malformed call, reconstructs the executable call, rewrites the retained history into canonical form, executes the action, and returns an explicit correction note. In the audited round, 405 sessions contain a recovery marker; each contains exactly one such marker and no logged recurrence after the correction. This is a system-counting fact. It does not show that weights learned the format or that recovery generalizes to other defects.

6 A Teacher-Forcing Failure at the Interface

Round 1 introduces a malformed parameter-key pattern that is absent in the base run. The emitted form places a Boolean value inside the parameter key, so the parser silently drops the intended argument. The training records contain the correct syntax, and the teacher favors the correct delimiter in clean base-policy contexts. A targeted two-sided probe, however, gives a different result after the malformed prefix is already present: at the divergence token, the 4B teacher assigns log probability -0.161 (about 85% probability) to continuing the malformed form, versus -0.253 under the student. On a matched correct-form state, the preference favors the correct continuation.

Thus teacher generation and teacher-forced grading are distinct objects. A teacher that generates canonical syntax from a clean history can locally reward continuation of a student's defect after that defect enters the context. Under dense on-policy distillation, this creates wrong-signed feedback at precisely the visited malformed prefix. We call this a *teacher-forcing copy-prior hypothesis*. The current evidence is one targeted defect family and a small probe, so the name describes a candidate mechanism rather than a general law.

This observation relates to exposure bias and to failures of token-level teacher agreement on degraded prefixes. It also gives a testable prediction: paired histories differing only in canonical versus malformed prior syntax should change teacher log-odds while holding state and candidate continuations fixed. Section 7 turns that prediction into a multi-state, multi-defect experiment.

7 Confirmatory Study Design

The repository contains a draft fail-closed launcher contract that enumerates the following matched arms. A hash-pinned preparation adapter validates and normalizes every arm's immutable record contract, but verified arm artifacts, several trainer-objective extensions, and accelerator execution do not yet exist. Results are not reported until those remaining implementation boundaries, the full protocol, and immutable run bundles exist.

Primary endpoint and unit. The independent unit is a six-hour run from a fresh canonical unseeded world. The preregistered primary endpoint is total CORE-3 stage gain across the three clustered agents. Quest-wall passage is secondary. All runs, confidence intervals, effect sizes, and stopping rules must be reported.

State-source ablation. The six primary training arms are natural OPD, targeted persistent player states, random-valid player states, progress-matched player states, TCOD-B2F teacher-success prefixes, and Guided-OPD. Four prespecified mechanism/baseline arms—visitation-only selection, teacher-advantage-only selection, corrected-interface SFT, and SCoRe-style first-error prefixes—are analyzed separately rather than silently counted as extra primary arms. Match teacher, optimizer, action-token and teacher-scoring budgets, environment interactions, and training-seed schedule; evaluate every arm from the original unseeded state. The method claim fails if random or progress- matched resets, TCOD-B2F, or Guided-OPD ties the targeted arm within the preregistered smallest effect of interest.

Player-state by history ablation. For the same certified player state, compare (i) a direct snapshot with the minimal canonical observation history, (ii) replay of a teacher trajectory with its authentic prefix, and (iii) a direct snapshot with a matched reconstructed history. A fourth Backplay condition anneals starts backward along a witness trajectory. This distinguishes initialization of latent player state x from information or error patterns carried in h, and tests whether a direct snapshot adds value beyond successful prefix replay. The present implementation does not restore a complete shared world (for example NPC, mob, resource, or concurrent-player state), so no full-world-state claim is made.

Weights by recovery factorial. Evaluate base, round-2, and round-3 weights with recovery both off and on. Preserve raw pre-rewrite emissions. This estimates main effects and interactions for stage progress and format-defect rate instead of attributing a combined system score to weights.

Generalization. Hold out at least one quest from candidate discovery, state selection, and training, then compare full-walkthrough against no-walkthrough prompts. Report the probability of reaching the bottleneck, crossing it conditional on arrival, and finishing downstream. A second environment or model family is needed for any broad claim. Corrected-interface SFT and a SCoRe-style first-error-prefix condition are additional baselines under matched interaction and teacher- scoring budgets (Lyu et al., 2025).

Copy-prior test. Construct paired canonical/malformed histories across tools, states, defect types, and at least two teacher sizes or families. Score identical canonical and malformed continuations; report teacher and student log-odds with paired confidence intervals. Then intervene on grading context and test generation, rather than inferring generation from grader-side movement.

8 Related Work

GKD establishes on-policy distillation from student-generated sequences with teacher feedback (Agarwal et al., 2024); our work does not claim novelty for OPD itself. OPCD distills contextual experience on student-generated trajectories, including in text games (Ye et al., 2026); On-Policy Expert Corrections switch from student to expert within a trajectory to address covariate shift (Lauffer et al., 2025). TCOD directly studies temporal curricula and intermediate-state initialization for multi-turn OPD (Wang et al., 2026); broad novelty for seeded starts is therefore preempted. Guided-OPD changes live occupancy by mixing teacher and student turns (Li et al., 2026); ReOPD designs reliable prefix distributions offline (Liao et al., 2026); and SCoRe localizes the first student error before short-horizon training (Lyu et al., 2025). SOD and KAT instead alter or terminate supervision on degraded prefixes (Zhong et al., 2026; Xin et al., 2026). These methods make intermediate-state, student-failure, and state-centric novelty claims unavailable. DAgger formalizes iterative data collection under the learner's state distribution (Ross et al., 2011). A recent state-distribution analysis likewise argues that the source and locality of training states can matter as much as the loss (Nie, 2026). Reset-along-demonstration, Backplay, and Go-Explore show that restoring intermediate states can expose useful signal in hard-exploration reinforcement learning (Salimans and Chen, 2018; Resnick et al., 2018; Ecoffet et al., 2021). We use the same control point for on-policy language-model distillation in a persistent, typed environment, but the current study does not establish superiority over those curriculum families.

Game-agent benchmarks already cover long-horizon language and vision agents. BALROG emphasizes knowing–doing gaps across games (Paglieri et al., 2025), while Orak uses MCP across twelve games and releases gameplay fine-tuning data (Park et al., 2026). MCP and game play are therefore infrastructure here, not novelty claims. Voyager and AgentTrek use curricula, reusable skills, or guided replay to obtain long-horizon behavior (Wang et al., 2024; Xu et al., 2025). Our narrower question is whether a frozen selector over low-occupancy, teacher-competent persistent player states adds value beyond natural OPD, successful-prefix replay, turn-level guidance, and matched generic resets.

Supervised imitation can compound errors under distribution shift (Ross et al., 2011), and knowledge distillation does not guarantee functional agreement between teacher and student (Stanton et al., 2021). The r10 cross-vocabulary SFT regression in our development record is motivating evidence, but its raw artifact bundle is incomplete and it is not a primary result of this submission draft.

9 Conclusion

The audited case supports a concrete hypothesis: low-occupancy player states may hide useful teacher competence from natural OPD. Natural-visitation OPD changed behavior without changing score; a later hand-selected state-augmented round was followed by unseeded passage of a previously unseen wall. That sequence does not establish the proposed selector. Random and progress-matched resets, TCOD-B2F, Guided-OPD, replication, and held-out transfer decide whether the narrow method survives. A malformed-prefix probe separately shows why teacher- forced token feedback can fail at the tool boundary.

Limitations

The central limitation is statistical: one run per principal arm cannot support a population claim, and three prompt variants are not independent seeds. The state-seeded round also differs from the preceding round in initialization and data. The environment is a single game with procedural walkthroughs, the teacher and student are from one model family, and stage counts are coarse. Some development analyses depend on logs whose raw inputs are not yet packaged. The paper must not be submitted as confirmatory until the study design above is run and a clean clone regenerates every main table.

Direct database construction is environment-specific and may not transfer to systems without restorable state. The historical failure state was selected retrospectively, so selection bias is possible; the prospective selector has no outcome yet. Teacher competence at the target state is observed in a small number of trajectories, not estimated precisely. The syntax probe covers one defect family and should not be generalized to all teacher-forced grading.

Ethical Considerations

The work concerns game agents acting in isolated local worlds. Risks include overstating autonomy, silently baking procedural knowledge into prompts, and mistaking recovered actions for model competence. We mitigate these by labeling scaffolding, logging model-visible schemas and raw emissions, separating runtime recovery from weights, and reporting failed interventions. No human-subject or private-user data are used.

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A Training Details

The teacher is a scaffolded Qwen-family 4B model and the student begins from a 2B checkpoint. Both use the same vocabulary. Student rollouts are reconstructed as the byte-exact serving prompt plus the verbatim emitted continuation. This prefix property avoids re-rendering differences caused by chat-template handling of reasoning delimiters. Teacher and behavior log probabilities are computed on the identical raw string. Every tenth session is held out for development gates.

The update uses PPO-style clipped importance weighting with clip $\epsilon = 0.3$, advantage clamp $[-3, 3]$, and a 1.5 weight on the first third of each session. Training uses one epoch, learning rate 5×10^{-5}, effective batch size 32, and a fresh LoRA initialized over the prior merged checkpoint. Action positions alone receive the language-model head. These choices were development decisions, not independently ablated.

Round 1 contains 5,564 train and 574 held-out records from the base policy's natural rollouts. Round 2 combines 5,523 natural round-1 records and 2,326 seeded records before the 7,024/825 train/held-out split. The seed encodes accepted Herbalist stage 1, Foraging level 5, starter inventory, and a verified tile near the target resource. Evaluation never loads this state.

B Claim-to-Evidence Boundary

Round-1 behavior. Descriptive, one run: measured behavior changed while score did not.

State-seeded round. Unreplicated: the seeded-training round was followed by unseeded wall passage in one run.

Targeted selector. Protocol only: no candidate-state bundle, trained checkpoint, or outcome is currently available.

12 to 15. Reported recovery-off historical contrast; not an effect estimate or clean weights comparison, and exact parity remains unavailable.

18/30. Weights-plus-interface system result only.

Copy prior. Context-dependent endorsement reversal for the targeted historical probe.

Recovery. No repeat marker after correction in rewritten logs; raw relapse is unavailable.

Cross-task transfer. Unsupported; no claim.

C Reproducibility Contract

Before confirmatory runs, each immutable run bundle must contain: environment and game revisions; container or dependency locks; model, tokenizer, adapter, and dataset revisions; complete model-visible tool schemas; rendered-prompt hashes; sampling parameters; environment and inference seeds; raw emissions before any rewrite; state-transition logs; analysis outputs; and checksums. Dataset records must point to source runs and transformations. A clean-clone command must verify hashes and regenerate every table and figure without network-dependent mutable inputs.

The current development artifact does not yet meet this bar. In particular, raw inputs for the historical r10 analysis and some OPD gate summaries are absent, and the dedicated r10 eval path reset a different database from its game servers after April 25. The June OPD runs used a separate database-aligned orchestrator lane, but their raw bundles are also absent. Full dependency installation has now been demonstrated for the prospective CPU unit suite in a fresh clone, but not for training, live game execution, checkpoints, or paper-table regeneration. The prospective launcher can seed inference. A draft game-side patch centralizes gameplay randomness and publishes a strict startup attestation, and the launcher verifies a distinct registered environment seed for each replicate. That patch is not yet merged, deployed, or verified in a live run, so confirmatory launch remains fail-closed. A draft isolated-service reachability checker executes hash-pinned MCP transitions and persistent player-state comparisons, but it has not yet produced a certificate in a live game/Mongo lane. A draft matched-training launcher freezes 50 core arm–seed cells plus 20 separately reported state–history ablation cells, shared budgets, interfaces, and held-out exclusion. A stacked draft adapter hash-verifies source material, checks held-out and arm-specific evidence, and emits normalized records marked `prepared_not_trained`; verified training bundles and the Guided, corrected-SFT, and SCoRe trainer extensions remain absent. These omissions are release blockers, not appendix footnotes.

D Planned Statistical Analysis

No session- or agent-level pseudo-replication will be used for policy effects. The run is the analysis unit. Before confirmatory outcome collection, the team must preregister a smallest effect of interest and determine the run count from a documented power calculation using independent pilot variance or a conservative prespecified variance grid. If variance is not defensibly known, the protocol must use blinded variance re-estimation with a fixed maximum sample size; there is no default "five runs per arm." The primary comparison is two-sided unless a directional alternative and stopping rule are preregistered before collection. We will report all runs, bootstrap or randomization-based confidence intervals appropriate to the achieved sample, standardized and raw effect sizes, and the full family of secondary endpoints with multiplicity handling. Exact categorical wall-passage counts will be reported descriptively when sample sizes are small.